

INTRODUCTION

TO

AUTIRONICS

MCC127

LECTURE NOTES

Pt

COMMUNICATION SIGNALS

Communication signals: - A computer uses

Voltage values as communication signals, thus voltage is often referred to as a signal or a voltage signal. There are two types of voltage signals

1. Analog
2. Digital

1. An Analog signal's voltage is continuously variable within a given range and time is used for the voltage to change. An analog signal is generally used to convey information about a medium that changes gradually and continuously within an extended period. Temperature sensing devices usually give off an analog signal. A digital signal's digital signals also vary but are not continuously, and time is not needed for the change to occur. Turning a switch on and off creates a digital signal, a voltage is either there or it is not. Digital signals are often referred to as square wave signals.

BINARY CODE: A computer converts a series of digital signals to a binary number made up of ones and zeros, voltage above a given threshold converts to 1 and voltage below that converts to 0. Each '1' or '0' represents a bit of information, eight bits equal a byte (sometimes referred to as a word).



DIGITAL BINARY SIGNAL

1.0 Micro Processor

Contrasting with a Microprocessor is a computer, which is a Microcomputer, fabricated in a single chip capable of performing Arithmetic logical (ALU) unit operations and communicating with other devices connected to it.

Most automotive computers have 8-bit Microprocessors, but as the computer takes on more responsibility and have to communicate more information more rapidly, 16-bit units will become appropriate.

1.1 INTERFACE: The Microprocessor or CPU is the heart of a computer, but it requires several support function, one of which is its interface. A computer has an input and an output interface circuit. The interface has two functions:

① It protects the delicate electronics of the Microprocessor from the higher voltages of the circuits attached to its computer, and it translates input and output signals.

The input interface translates all analog input data to binary code. Most sensors produce an analog signal. It is sometimes referred to as A/D, analog to digital. The output interface, D/A, translates digital signals to analog form and provides function that need an analog voltage.

1.2 Microcomputer: The microprocessor of a small computer does its calculations and makes several other information. The computer is therefore equipped with information storage capability called memory.

The computer actually has three memories:

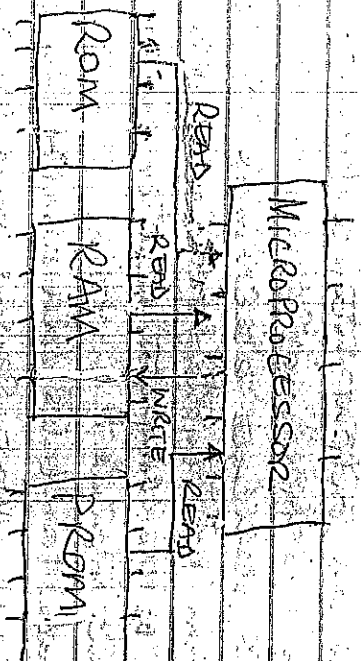
1. Read-only memory (ROM)
2. Programmable read-only memory (PROM)

pg

3. Random access memory (RAM).

The ROM contains permanently stored information that instructs (programs) the microprocessor on what to do in response to input data. The microprocessor can read information from the ROM but cannot put information into it. The ROM unit is soldered into the computer and is not easily removed.

The PROM differs from the ROM in that it plugs into the computer and is more easily removed and reprogrammed or replaced with one containing a revised program. It contains program information specific to different vehicle model calibrations.



The three memories within a computer

The RAM is where temporary information is stored. It can be both read from and written into it. For example, data comes in from a sensor and must be used for several different decisions. Such as manifold vacuum information. The microprocessor will send it in RAM until it is up dated.

In most automotive computers, the computer test at least some of the input and output signals to see whether the circuit that sent the signals is received. The command is working.

P₄

properly. The computer knows what the voltage will be if the circuit were open and what the voltage will be if it were shorted. It watches to see that the voltage remains between these voltage values. On some circuits, the voltage sensor circuit from instance, the computer expects to see the voltage go up and down, crossing a certain value. If during its continuous testing it sees something wrong (a fault), it will record information in RAM concerning the fault. The RAM is also labeled in place and not easily removed.

There are two kinds of RAMs.

1. Volatile

2. NonVolatile

1. A Volatile RAM is sometimes called RAM, the keep-alive memory. It must have a constant source of voltage to continue to exist and is erased when disconnected from its power source. In automotive applications, a volatile RAM is usually connected directly to the battery by way of a fuse or fusible link so that when the ignition is turned off, the RAM is still powered.

2. A nonVolatile RAM

Stores information of its power source is disconnected. Vehicles with digital display odometers usually store mileage information in a nonvolatile RAM.

Data links — When computers communicate with other electronic devices such as control panel, modules, some sensors, or other computers in the form of digital signals, they communicate.

Through Corts called data links. Some data links transmit data in only one direction, although other transmit in both directions. What makes a data link different from an ordinary circuit? The manner in which it is controlled. Some control must be used so that its devices at each end or its data link know when to transmit and when to receive.

QUESTIONS (REVIEW)

1. Name two types of Voltage Signal, give me example for each.
2. What does its abbreviation ROM stand for, and what is its function?
3. Name at least two ways that a PROM is different from a ROM.
4. Define Microprocessor.
5. What is data link?
6. What makes data link different from an ordinary circuit? In an Automobile telephone system.

Ps

SENSOR

SENSOR:- A sensor is an electronic device that monitors various aspect of the vehicle engine and sends information to the driver or Electronic Control Unit (ECU).

Sensors can monitor various aspect of a car such as its temperature, coolant, by slams, engine oil pressure, emission levels, vehicle speed, and levels are the best examples to understand the working principle of sensors. They send signals to the brain which then makes a decision. Similarly, car sensors send signals to the ECU to make appropriate adjustment to run the engine.

TYPES OF CAR SENSORS

The table below shows different types of sensors in a car and their functions.

Sensor	Function
Air-Flow Sensor	It measures the density and volume of the air entering the combustion chamber
Engine Knock Sensor	It monitors engine knocking and ensures its air-fuel mixture is limited correctly
Engine Speed Sensor	It monitors the spinning speed and position of the Crankshaft
Camshaft Position Sensor	It monitors its position and prevents timing of the Camshaft
Manifold Absolute Pressure (MAP) Sensor	Monitors engine load by measuring the difference between the manifold and outside pressure
Throttle Position Sensor	Monitors the position of the throttle valve

Pg

Voltage Sensor

It manages the locking speed of the vehicle.

Oxygen Sensor

It helps to measure the oxygen level present in the exhaust gases.

NO_x Sensor

It measures the oxides of Nitrogen (NO_x) present in the exhaust gases.

Temperature Sensor

It monitors the engine temperature.

Fuel Temperature Sensor

It monitors the temperature of the fuel entering the engine.

Speed Sensor

It measures the speed of the wheels.

Parking Sensor

It recognises any obstacle present in the front of the vehicle.

Run Sensor

It detects rain and sends a signal to ECU to activate the wiper.

P8

1. Air-Flow Sensor

- It is a type of sensor that measures the volume and density of air entering the engine.
- It is placed inside the air-conditioner chamber.
- It ensures the right amount of air-fuel mixture enters the combustion chamber for optimum performance.
- Failure of the air-flow sensor can result in engine stalling or more fuel consumption.

2. Engine Knock Sensor

- Knocking is a phenomenon where the air-fuel mixture ignites before the spark plugs produce the spark.
- It reduces engine efficiency, and in its long run, it can cause damage to its engine components.
- The engine knock sensor detects knocking and monitors the air-fuel mixture ignition process.
- The sensor can warn the driver of knocking or fault in the ignition of the air-fuel mixture.

3. Engine Speed Sensor

- It is a sensor that monitors the speed and position of the crankshaft.
- The sensor is fitted to the crankshaft.
- It detects the crankshaft position and relays the information to the ECU.

- The ECU then calculates the exact time for fuel injection and adjusts the spark plug.

- If the timing is not correct, the engine loses its efficiency.

- It is one of the most important sensors in a car.

4. Crankshaft Position Sensor

- A crankshaft controls the inlet and exhaust valve of an engine.
- The crankshaft position sensor monitors the

P₇

Timing of the Crankshaft so that the air enters the cylinders and burnt gases are sent out of the cylinder at the right time.

- If the timing is off, it hampers the power output, and in the worst case it may damage the cylinder.

5. Manifold Absolute Pressure (MAP) Sensor

- It is a sensor that monitors the engine load.
- It calculates the difference between the manifold and outside pressure.

- The MAP sensor ensures that the engine receives fuel based on the change in pressure.

- The failure of the MAP sensor may result in high fuel consumption.

6. Throttle Position Sensor

- It measures the position of the throttle valve when you press the accelerator pedal.

- The sensor sends the information to the ECU, which then decides how much air-fuel mixture needs to be sent to the cylinder.

- A faulty throttle position sensor hampers the power delivery.

7. Voltage Sensor

- The voltage sensor is responsible for measuring the vehicle's idling speed.

- The sensor also ensures that idling speed is decreased or increased when required.

8. Oxygen Sensor

- As its name suggests, the oxygen sensor measures the amount of oxygen present in the exhaust gases.

P
10

- Generally, the O₂ sensor is located near the exhaust manifold and after the Catalytic converter.
- The sensor helps determine indicator, the engine is burning a rich or lean air-fuel mixture.
- A faulty oxygen sensor can cause high fuel consumption, jacking or a problem with idling speed.

9. NOx Sensor

- The NOx sensor monitors the oxides of Nitrogen (NOx) present in the exhaust gases.
- There are strict laws in respect to the permissible level of NOx emission from a vehicle.
- Hence, the NOx sensor is essential to monitor the emission of harmful gases.
- The sensor will relay the information to the ECU, which then makes adjustments to the air-fuel mixture to achieve the required NOx levels.

10. Temperature Sensor

- It is a sensor that can monitor the temperature of the engine.
- Thermal management is critical for the smooth running of a vehicle. Hence, it is one of the critical sensors in a car.
- The sensor is also responsible for the efficient functioning of the cooling system.
- The sensor can also help to warn the driver if the engine is running hot or coolant level is low, which is not working correctly.

11. Fuel Temperature Sensor

- It is a sensor that constantly monitors the fuel temperature to ensure that fuel consumption is optimum.

P₁₁

- If the fuel is cold it takes more time to burn due to its high density. Similarly, if the fuel is warm, it ends up burning its particles.
 - The fuel consumption sensor ensures that the fuel is supplied to its cylinder at its optimum temperature for its smooth running of the engine.
- ### 12. Speed Sensor
- As the name suggests, it monitors the speed of the car's wheels.
 - A car's wheels can travel at different speeds when going around a corner or while off-roading.
 - Hence, the speed sensor helps to brake the speed of each wheel.
 - The data is sent to the ECU, and it can be utilized by safety features such as traction control, ABS (Anti-lock Braking System).
 - The speed sensor is crucial, especially for 4x4 vehicles.

ADVANTAGES OF SENSORS

- Car sensor maintenance is very easy.
- The sensor can easily detect faulty components in a vehicle.
- Sensors ensure that the engine is maintained properly.
- Sensors also enable automatic control of specific functions such as windshield wipers, headlights, etc.
- The ECU can make precise adjustments with the information received from sensors.
- Sensors can also relay warning information to the driver of issues in any fault/malfunction with the car's components.

P
r

DISADVANTAGES OF SENSORS

- One major disadvantage is having so many sensors in board in that they can fail over time.
- A faulty sensor can lead to damage to internal components of the vehicle. Getting things repaired or replaced can be an expensive affair.

QUESTION

1. What is a sensor?
2. In a tribution firm, state its functions.
3. What happens if the Oxygen sensor is not working correctly?
 - If the oxygen sensor is a fault, the ECU will not get the information regarding the air-fuel mixture ratio that is supplied to the engine. It can also cause idling problems, jerkiness or issues with power delivery.

ACTUATORS

Actuators are essential part of electronic control systems in passenger cars that convert the electrical signals from the control unit into an action. Most actuators are electro-motors or electro-magnetic valves.

The computer uses the sensor data to control different systems inside vehicle through the use of actuators.

Definition: An actuator is an electro-mechanical device that can convert electrical signals from the control unit into an action such as a relay, solenoid or motor. Actuators can adjust engine speed, change suspension height or regulate the fuel mixture into the engine.

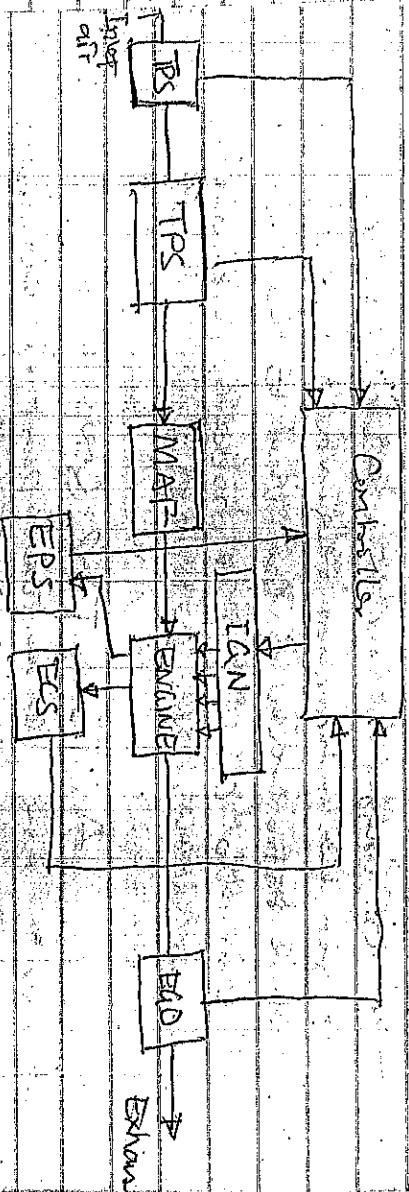
P13

ENABLE MAINTENANCE SYSTEM (EMS)

1. ELECTRONIC FUEL CONTROL SYSTEM

The primary function of this fuel control system is to accurately determine the mass of air flow rate into the engine. The control system precisely regulates fuel delivery such that the ratio of its mass of air to the mass of fuel in each cylinder is a close as possible to stoichiometry (i.e. 14.7).

For an understanding of the configuration of electronic fuel control system, refer to the block diagram below.



The components of the block diagram are as follows:-

1. Throttle Position Sensor (TPS)

2. Mass Air Flow Sensor (MAF)

3. Fuel injectors (FI)

4. Ignition System (IGN)

5. Exhaust gas Oxygen Sensor (EGO)

6. Engine Control Sensor (ECS)

7. Engine Position Sensor (EPS)

The EPS has the capability of measuring crankshaft angular speed (RPM) as well as crankshaft angular position when it is used in conjunction with a stable and precise electronic clock (in its controller).

The signals from the various sensors enable the controller to determine its correct fuel flow in relation to its air flow to obtain the stoichiometric mixture. From this calculation the correct fuel delivery is regulated via fuel injectors. In addition, optimum timing is determined and appropriate timing pulses are sent to the ignition control module (ICM).

OPEN LOOP CONTROL

The components of an open-loop controller include the electronic controller which gives an output to an actuator.

The actuator, in turn, regulates the plant being controlled in accordance with the desired relationship between its reference input and its value of the controlled variable in its plant.

Many examples of open-loop control are encountered in automotive electronic systems, such as fuel control in carburetor operating modes. In its open-loop control system of fig 2 below, the command input is sent to a system block, which performs the control operation on the input to generate an intermediate signal that drives the engine.

This type of control is called open-loop control because its output of the system is never compared with the command input to see if they match. The control electronics generates the electrical signal for the actuator in response to its control input and in accordance with the desired

P15

Relationship between the control input and the system output

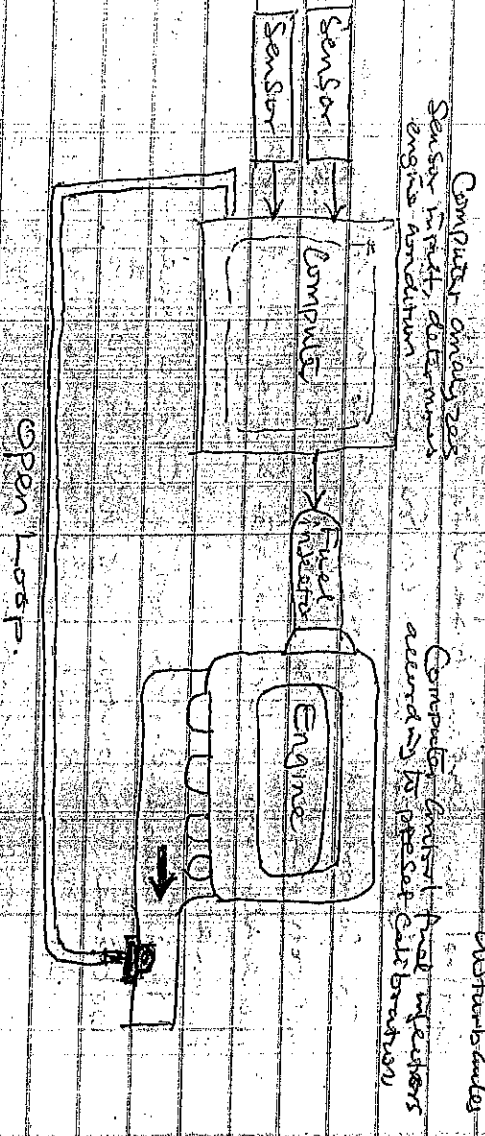
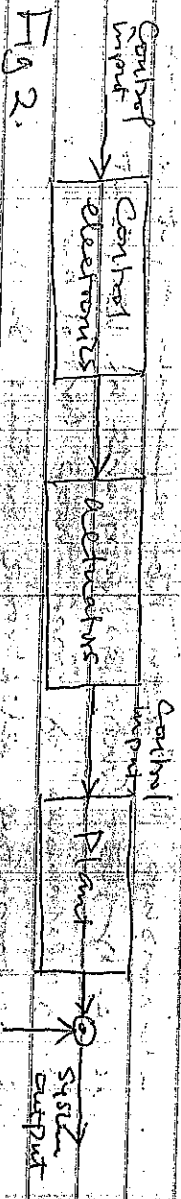
The operation of the plant is directly regulated by the actuator (which must supply the an electric motor).

The system output may also be affected by external disturbances that are not an inherent part of the plant but as the result of the operating environment.

DISADVANTAGE OF OPEN-LOOP CONTROL

One of the principal drawbacks to the open-loop controller is its ability to compensate for changes that might occur in the controller or the plant or for any disturbances.

This defect is eliminated in a closed-loop control system, in which the actual system output is compared to the desired output value is accelerated into its input.



File

During engine cranking the machine is set with an amount depending on the engine temperature (measured via the engine coolant sensor).

Once the engine starts and until a specific set of conditions is satisfied, the engine control operates in its open-loop mode, but in mode the mass air flow is measured (via MAF sensor).

The correct fuel amount is computed in the electronic controller as a function of engine temperature.

The correct actuating signal is then computed and sent to the fuel metering actuator.

In essentially all modern engines, fuel metering is accomplished by a set of fuel injectors. After combustion the exhaust gases flow past the EGR sensor through the IMC, and out the tailpipe.

Once the EGR sensor has reached its operating temperature (typically a few seconds at about 2 min) the EGR sensor signal is read by the controller and the system begins closed-loop operation.

CLOSED-LOOP CONTROL

In a closed loop control system a measurement of the output variable being controlled is obtained.

Via a sensor and fed back to the controller.

The measured value of the controlled variable is compared with the desired value for that variable based on the reference input.

An error signal based on the difference between desired and actual values of the output signal is

generated, and the controller generates an actuating signal that tends to reduce the error to zero.

P₁₇

in addition to reducing the error to zero feedback has other potential benefits in a control system.

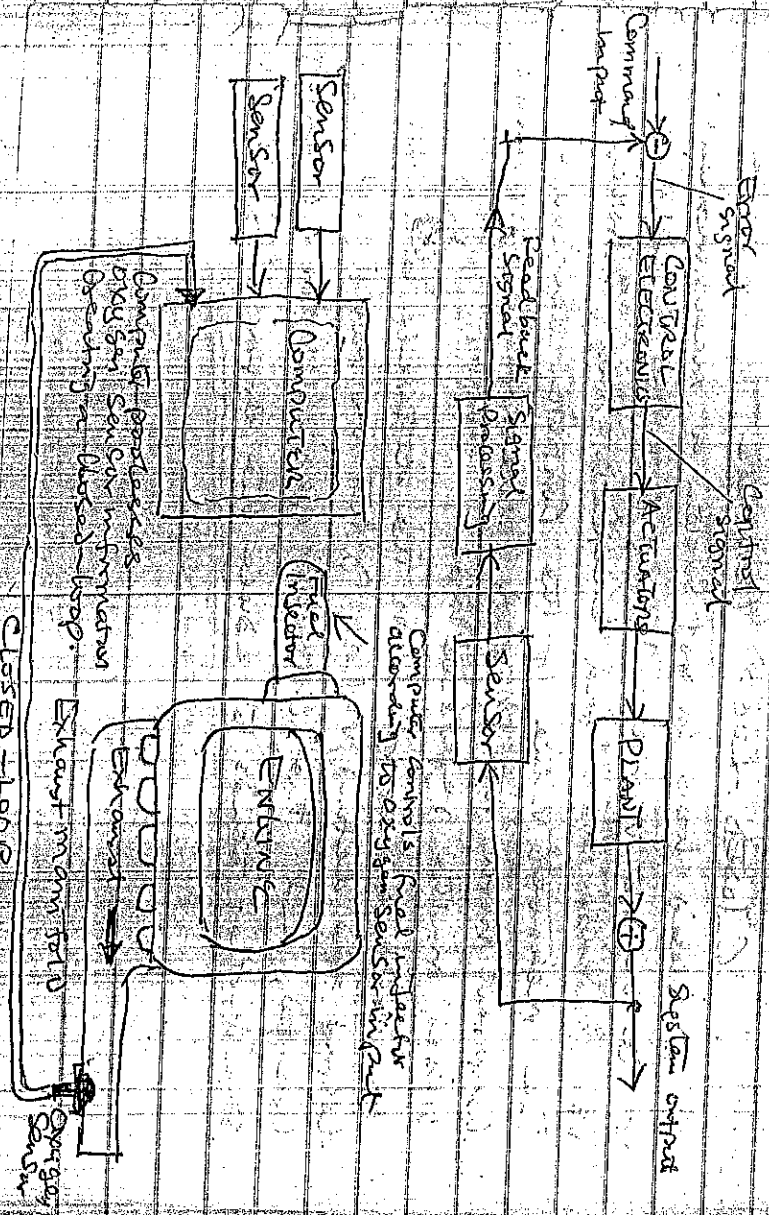
ADVANTAGES OF FEEDBACK

- It can affect control system performance by improving system stability and suppressing the effects of disturbances in the system.

For any given set of operating conditions, the fuel metering actuator provides fuel flow to produce an air/fuel ratio set by the controller input.

The mixture is burned in the cylinder and the combustion products leave the engine through the exhaust pipe.

The EGO sensor generates a feedback signal for the controller input that depends on the air/fuel ratio. This signal tells the controller to adjust the fuel flow rate for the required air/fuel ratio, thus completing the loop.



P18.

QUESTIONS

1. ~~What~~ What is its primary function of Electronic Fuel Control System?
2. Write its need of block diagram, outlined its components & its electronic fuel control system (EENS).
3. What is the difference between its open-loop control system and closed-loop?
4. State the advantages of feed back in a closed-loop system engine control system.
5. Use block diagram to explain its difference between open loop engine control system and closed-loop engine control system.